How to install and configure claude code (CLI) on macOS

For ***non-developer users***, there is: [**Claude Code (1-Click)(brew)**](jamfselfservice://content?entity=policy&id=1027&action=view)

- The biggest difference is it enforces “claudeai” authentication tied to our Claude Org ID via a managed-settings.json

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Overview

This guide walks you through installing and configuring Claude Code, a command-line interface (CLI) tool, on your macOS device. Because our environment authenticates Claude via Google's Vertex AI, the setup requires installing a few foundational pieces first. Specifically, Vertex AI requires the `gcloud` tool, which strictly depends on specific versions of Python.

To make this seamless for non-developers, we heavily rely on the Self Service application to handle the complex installations automatically.

The installation is broken down into four main phases:

- **Phase 1: Prerequisites:** You will configure your environment for Zscaler Internet Security and install Homebrew (a package manager) using Jamf Self Service.
- **Phase 2: Python Setup:** You will install `pyenv` via Self Service to download and manage the exact version of Python required by Google Cloud. You will also configure your `~/.zshrc` file to ensure your Terminal recognizes it.
- **Phase 3: Google Cloud Setup:** You will download the necessary Python version, create a virtual environment, and install the Google Cloud SDK (`gcloud`) via Self Service.
- **Phase 4: Claude Installation:** You will add the final environment variables (like your Vertex Project ID) to your `~/.zshrc` file and install the Claude Code tool via Self Service.

Phase 1: Prerequisites

Configure shell environment to play nicely with Zscaler Internet Security (ZIA).

- **Self Service:** [Zscaler (Shell Environment)](jamfselfservice://content?entity=policy&id=1321&action=view)

Install Homebrew, a package manager which facilitates the installation of many tools/applications.

- **Self Service:** [Homebrew (managed)](jamfselfservice://content?entity=policy&id=967&action=view)

Force a check-in with the MDM (a.k.a Jamf), this should help ensure that the MDM is aware of the recent Homebrew install and make brew-related Self Service items available.

- **Self Service:** [Jamf Check for Updated Policies](jamfselfservice://content?entity=policy&id=144&action=view)

Phase 2: Python Setup

- Launch Self Service
- Run the item: [Update and Upgrade (brew)](jamfselfservice://content?entity=policy&id=993&action=view)
- Run the item: [Choose Your Own Brew](jamfselfservice://content?entity=policy&id=982&action=view)
  - When prompted, enter: `pyenv` in the dialog.
- Configure `zsh` startup file(s) for `pyenv`:

> pyenv zsh startup file configuration
>
> Open "Terminal.app" and run: `open -e ~/.zshrc`
>
> and add the following to the file:
>
> ```
> # pyenv config
> export PYENV_ROOT="$HOME/.pyenv"
> [[ -d $PYENV_ROOT/bin ]] && export PATH="$PYENV_ROOT/bin:$PATH"
> eval "$(pyenv init - zsh)"
> ```
>
> Save
>
> Quit and relaunch Terminal

Install the `pyenv-virtualenv` plugin:

> commands to install pyenv-virtualenv plugin
>
> Open "Terminal.app" and run:
>
> ```
> mkdir -p ~/.pyenv/plugins
> cd ~/.pyenv/plugins
> git clone https://github.com/pyenv/pyenv-virtualenv.git
> ```

Configure `zsh` startup file(s) for `pyenv-virtualenv`

> pyenv-virtualenv zsh startup file configuration
>
> Open "Terminal.app" and run: `open -e ~/.zshrc`
>
> and add the following to the file:
>
> ```
> # pyenv-virtualenv config
> eval "$(pyenv virtualenv-init -)"
> ```
>
> Save
>
> Quit and relaunch Terminal

Download version of Python needed for gcloud

- See "Depends on" at: https://formulae.brew.sh/cask/gcloud-cli#default
- As of 2026.02.20, it requires: `3.13.12`
- Run: `pyenv install 3.13.12` to download Python.

Create a Python virtual environment for gcloud

- Run: `pyenv virtualenv 3.13.12 gcloud-env`

Configure `gcloud` to use this Python virtual environment:

> gcloud zsh startup file configuration

Open “Terminal.app” and run: `open -e ~/.zshrc`

and add the following to the file:

```
# gcloud config
export CLOUDSDK_PYTHON="$HOME/.pyenv/versions/gcloud-env/bin/python"
```

Save

Quit and relaunch Terminal

Phase 3: Google Cloud Setup

Install gcloud

- Launch Self Service
- Run the item: [Google Cloud SDK (brew)](jamfselfservice://content?entity=policy&id=985&action=view)
  - **Please note:** This item commonly reports “failed”. It is usually a false alarm. Quit and relaunch Terminal and run: `gcloud --version`, if a version is displayed then all is well and the “failed” notice can be ignored.

more gcloud zsh startup file configuration

Open “Terminal.app” and run: `open -e ~/.zshrc`

and add the following to the file:

```
# gcloud config
source "$(brew --prefix)/share/google-cloud-sdk/path.zsh.inc"
source "$(brew --prefix)/share/google-cloud-sdk/completion.zsh.inc"
```

Save

Quit and relaunch Terminal

Run: `gcloud auth application-default login`

What if the gcloud install actually fails?

If `gcloud` actually fails to install....

Some users have had success uninstalling brew’s `openssl@3`

- **Note:** from Self Service, run: “**Choose Your Own Brew (Uninstaller)**” entering `openssl@3` when prompted.

Then retry the gcloud Self Service item.

Phase 4: Claude Installation

Configure `zsh` startup file(s) for `claude`

claude zsh startup file configuration

Open “Terminal.app” and run: `open -e ~/.zshrc`

and add the following to the file:

```
# claude config
export ANTHROPIC_VERTEX_PROJECT_ID='ss-p-inf-aihub-cp1t'
export CLAUDE_CODE_USE_VERTEX=1
export CLOUD_ML_REGION='us-east5'
export CLAUDE_CODE_DISABLE_NONESSENTIAL_TRAFFIC=1
```

Save

Quit and relaunch Terminal

Install claude

- Launch Self Service
- Run the item: [Claude Code (brew)](jamfselfservice://content?entity=policy&id=1473&action=view)
- Verify claude is installed by running: `claude --version`
- (optional) See the CLAUDE.md at: https://github.com/b1-as/bullish-ai/tree/master

Appendix

A. (Optional) Manual Install Instructions (Advanced)

(optional) manual install of pyenv

⚠ Prefer the Homebrew install of pyenv.

`pyenv-virtualenv` is automatically installed when `pyenv` is installed in this manner.

```
cd "${HOME}"
curl -L https://github.com/pyenv/pyenv-installer/raw/master/bin/pyenv-installer | bash
echo '# pyenv config' >> ~/.zshrc
echo 'export PYENV_ROOT="$HOME/.pyenv"' >> ~/.zshrc
echo '[[ -d $PYENV_ROOT/bin ]] && export PATH="$PYENV_ROOT/bin:$PATH"' >> ~/.zshrc
echo 'eval "$(pyenv init - zsh)"' >> ~/.zshrc
echo 'eval "$(pyenv virtualenv-init -)"' >> ~/.zshrc
source ~/.zshrc
pyenv install 3.13.12
pyenv virtualenv 3.13.12 gcloud-env
```

(optional) manual install of gcloud

⚠ Prefer the Homebrew install of gcloud.

gcloud's `install.sh` will ask to install Python. Select "**No**" since that has been taken care of already by pyenv-virtualenv. Select "**Yes**" when prompted to configure PATH.

```
cd "${HOME}"
curl -O https://dl.google.com/dl/cloudsdk/channels/rapid/downloads/google-cloud-cli-darwin-arm.tar.gz
tar -xf google-cloud-cli-darwin-*.tar.gz -C ~/
rm -rf google-cloud-cli-darwin-arm.tar.gz
export CLOUDSDK_PYTHON="$HOME/.pyenv/versions/gcloud-env/bin/python"
echo '# gcloud config' >> ~/.zshrc
echo 'export CLOUDSDK_PYTHON="$HOME/.pyenv/versions/gcloud-env/bin/python"' >> ~/.zshrc
cd ~/google-cloud-sdk
./install.sh
gcloud auth application-default login
```

(optional) manual install of claude

⚠ Prefer the Homebrew install of claude. *(might only work in certain geographical locations)*

```
cd "${HOME}"
curl -fsSL https://claude.ai/install.sh | bash
echo '# claude config' >> ~/.zshrc
echo 'export PATH="$HOME/.local/bin:$PATH"' >> ~/.zshrc && source ~/.zshrc
```

B. Helpful Links

Claude

- https://code.claude.com/docs/en/setup
- https://code.claude.com/docs/en/google-vertex-ai

GCloud

- https://docs.cloud.google.com/sdk/docs/install-sdk
- https://formulae.brew.sh/cask/gcloud-cli#default

Pyenv / Python

- https://github.com/pyenv/pyenv
- https://formulae.brew.sh/formula/pyenv#default
- https://github.com/pyenv/pyenv-virtualenv
- https://www.python.org/downloads/

Bullish

- https://github.com/b1-as/bullish-ai/tree/master

C. Troubleshooting / FAQ

Claude "API error"

Likely need to run: `gcloud auth application-default login`

D. Glossary of Terms

- **Claude Code:** A command-line interface (CLI) tool that this guide helps you install and configure on your macOS device.
- **CLI (Command-Line Interface):** A text-based interface where you interact with tools by typing commands instead of clicking buttons.
- **Environment Variables:** Specific settings or values—like your Vertex Project ID—that are added to your configuration files.
- **gcloud (Google Cloud SDK):** A foundational tool required by Google's Vertex AI. It strictly depends on specific versions of Python to run.
- **Homebrew:** A package manager which facilitates the installation of many tools and applications on a Mac.
- **Jamf (MDM):** Mobile Device Management software that ensures your device is synced with company policies and makes Homebrew-related Self Service items available.
- **pyenv:** A tool used to download and manage the exact version of Python required by Google Cloud.
- **Python:** A programming language that serves as a strict requirement for the gcloud tool.
- **Self Service:** An application on your Mac that handles complex software installations automatically, making it seamless for non-developers.
- **Terminal:** The built-in macOS application ("Terminal.app") where you will run commands and open configuration files.
- **Vertex AI:** A Google platform that our environment relies on to authenticate Claude.

- **Virtual Environment:** An isolated Python workspace created specifically for gcloud.
- **~/.zshrc file:** A startup configuration file that ensures your Terminal recognizes the new tools you install.
- **Zscaler Internet Security (ZIA):** A security application that requires a quick environment configuration so your shell environment plays nicely with it.

E. Gemini "(gemini-cli)" Setup

Prerequisites: Complete phases 1-3 of the above guide.

Configure `zsh` startup file(s) for `gemini`

gemini-cli zsh startup file configuration

Open "Terminal.app" and run: `open -e ~/.zshrc`

and add the following to the file:

```
# gemini-cli config
export GEMINI_MODEL='gemini-3-pro-preview'
export GOOGLE_CLOUD_PROJECT='ss-p-inf-aihub-cp1t'
export OTLP_GOOGLE_CLOUD_PROJECT='ss-p-inf-aihub-cp1t'
export GOOGLE_CLOUD_LOCATION='global'
```

Save

Quit and relaunch Terminal

Install gemini-cli

- Launch Self Service
- Run the item: [Choose Your Own Brew](jamfselfservice://content?entity=policy&id=982&action=view)
  - When prompted, enter: `gemini-cli` in the dialog.
- Quit and relaunch "Terminal.app"
- Verify gemini-cli is installed by running: `gemini --version`

Configure gemini-cli settings

gemini-cli settings.json file

Open "Terminal.app" and run:

```
mkdir -p ~/.gemini
touch ~/.gemini/settings.json
open -e ~/.gemini/settings.json
```

and replace the contents of the file with the following:

```
{
  "usageStatisticsEnabled": false,
  "selectedAuthType": "vertex-ai",
  "telemetry": {
    "enabled": true,
    "target": "gcp"
  },
  "general": {
    "previewFeatures": true,
    "enablePromptCompletion": true
  },
  "security": {
    "auth": {
      "selectedType": "vertex-ai"
    }
  }
}
```

Run: `gcloud auth application-default login`

Misc

- (optional) See the GEMINI.md at: https://github.com/b1-as/bullish-ai/tree/master